Aarhus university

Computer-Science

Numerical Linear Algebra

Handin 6

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Math done by a human

Consider the space of differentiable functions $f : [0, \pi] \to \mathbb{R}$ with the L^2 inner product

$$\langle f, g \rangle = \int_0^\pi f(x)g(x)\,dx. \tag{1}$$

Showing Orthogonality

For the given collection of functions defined on $x \in [0, \pi]$

$$\sin(x), \sin(2x), \ldots, \sin(kx), \quad \text{For every integer } k > 0 \tag{2}$$

Using equation(1) and applying the collection(2), it's possible to express the inner product of collection as such:

$$\int_0^\pi \sin(nx)\sin(mx)\,dx \quad \text{for } m \neq n \tag{3}$$

note the $m \neq n$ is used since $\langle f, f \rangle$ is the norm.
This integral can be made simpler by using the trigonometric identity 9.15 [1].

$$\int_0^\pi \frac{1}{2}[\cos((n-m)x) - \cos((n+m)x)]\,dx \quad \text{for } m \neq n$$

then using the integral rules, by taking the constants out and splitting the integral.

$$\frac{1}{2}[\int_0^\pi \cos((n-m)x)\,dx - \int_0^\pi \cos((n+m)x)\,dx] \tag{4}$$

from here each integral can be evaluated, starting with the first one.
Considering the first integral from equation(4), $\cos((n-m)x)$, this can be seen as a composite function on the form $h(i(x))$, where $h(u) = \cos(u)$ and $i(x) = (n-m)x$.
This means the integral can be evaluated by substitution.
Let $u = (n-m)x$ and $du = (n-m)dx$, then express dx as a product of u.

$$du = (n-m)dx$$
$$dx = \frac{1}{(n-m)}du$$

Note the limits $\int_0^\pi$ also change to accomodate u, this is done by putting the value of the limits into $i(x)$, which gives.

$$i(\pi) = (n-m)(\pi) = \pi(n-m)$$
$$i(0) = (n-m)(0) = 0$$

then substitute the new limits into the respective integral that was chosen.

$$\int_0^{\pi} \cos((n-m)x)\,dx \to \int_0^{\pi(n-m)} \cos(u)\frac{1}{(n-m)}\,du$$

remember to take the constant out:

$$\frac{1}{(n-m)}\int_0^{\pi(n-m)} \cos(u)\,du \tag{5}$$

now to integrate and evaluate (5).

$$\begin{aligned}
&\frac{1}{(n-m)}[\sin(u)]_0^{\pi(n-m)} \\
=&\frac{1}{(n-m)}(\sin((n-m)\pi) - \sin(0)) \\
=&\frac{1}{(n-m)}(0-0) \\
=&0
\end{aligned}$$

The reason for this is that $n, m \in k$ (integers) and $\sin(\pi) = 0$. Since any whole multiplier of π is either a half (odd number) or whole (even number) rotation which, in Mr.sin() eyes, is still 0.
It's assumed sin(0) needs no explanation.

With all that in order, the same can be shown for the second integral in equation(4) resulting in:

$$\frac{1}{2}[0-0] = 0$$

which means the inner product of the collection(2) is 0 meaning it's an orthogonal collection.

Constant function $f(x)$

Given the constant funnction $f(x) \equiv 1$, how does that change the orthogonality when k is 'even' or 'odd'?.
Inserting the new $f(x)$ in equation(1).

$$\langle f, g\rangle = \int_0^{\pi} 1 \cdot g(x)\,dx.$$

this prevents the use of trigonominal identities resultig in evaluating cos() instead of sin() in the integral.

By evaluating the integral via substitution, like in the previous integral.

$$\begin{aligned}
&\int_0^{k\pi} \sin(u)\frac{1}{k}\,du \\
&= \frac{1}{k}\int_0^{k\pi} \sin(u)\,du \\
&= \frac{1}{k}[-\cos(u)]_0^{k\pi} \\
&= \frac{1}{k}(-\cos(k\pi) - (-\cos(0))) \\
&= \frac{1}{k}(\cos(0) - \cos(k\pi)) \\
&= \frac{1}{k}(1 - \cos(k\pi))
\end{aligned}$$

And now for the magic! $\frac{1}{k}(1-\cos(k\pi)) = \begin{cases} 0 & \text{for an even } k \\ \frac{2}{k} & \text{for an odd } k \end{cases}$ which means if k is odd it isn't orthogonal and if it's even it is orthogonal.
So for $\sin(x)$, $\sin(3x)$ isn't orthogonal and $\sin(2x)$, $\sin(4x)$ is orthogonal.

Projection

Determinign the projection onto $\sin(x)$ and $\sin(3x)$ using $f(x) \equiv 1$ yiels:

$$proj_{\sin(x)}(1) = \frac{\int_0^\pi 1 \cdot \sin(x)\,dx}{\int_0^\pi \sin(x)\cdot\sin(x)\,dx}\sin(x) \tag{6}$$

$$proj_{\sin(3x)}(1) = \frac{\int_0^\pi 1 \cdot \sin(3x)\,dx}{\int_0^\pi \sin(3x)\cdot\sin(3x)\,dx}\sin(3x) \tag{7}$$

Using the results from the previous section, the numerator in(6) and(7) can be generalized to $\frac{2}{k}$ since both $\sin(x)$ and $\sin(3x)$ use odd k values.
Before inserting anything the denominator shall be evaluated, starting with equation(6).

$$\int_0^\pi \sin(x)\cdot\sin(x)\,dx$$

The above integral can be simplified using the trigonometric identities 9.15 [1] but for $\alpha = \beta$ and $\alpha = x$

$$2\sin^2(x) = -\cos(2x) + \cos(0)$$

isolate $\sin^2(x)$ to fit the case, remember $\cos(0) = 1$

$$\sin^2(x) = \frac{1-\cos(2x)}{2} \tag{8}$$

substituting that into the integral

$$\int_0^\pi \frac{1-\cos(2x)}{2}$$

extracting the constant and evaluating gets the following

$$\frac{1}{2}[x - \sin(2x)]_0^\pi$$

when evaluated

$$\frac{1}{2}((\pi - 0) - (0 - 0)) = \frac{1}{2}\pi = \frac{\pi}{2} \tag{9}$$

It's important to note for this, that the result from integrating equation(8) using either $\sin^2(x)$, as was just done, or $\sin^2(3x)$ would yield the same result $\frac{\pi}{2}$ since $\sin(2x)$ and $\sin(6x)$ both equal 0.
By combining the findings for the denominator(9) and the numerator(see start of section **Projection**), the equations(6)(7) can be rewritten as:

$$proj_{\sin(x)}(1) = \frac{2}{\frac{\pi}{2}}\sin(x) = \frac{4}{\pi}\sin(x) \tag{10}$$

$$proj_{\sin(3x)}(1) = \frac{\frac{2}{3}}{\frac{\pi}{2}}\sin(3x) = \frac{4}{3\pi}\sin(3x) \tag{11}$$

these projections represents (mathematically) the place on the sine waves the looks like the line $f(x) \equiv 1$ the most. According to equation (13.3) [1] adding the projections together will result in a better approximation of line on the span.

The given $g(x) = 1 + c_1\sin(x) + c_3\sin(3x)$ looks a bit like the (c) from proposition 13.10 [1], the residual vector.

$$f(x) - (proj_{\sin(x)}(1) + proj_{\sin(3x)}(1)) \perp w \tag{12}$$

and remember $f(x) \equiv 1$, so the difference is that $g(x)$ adds the projected vector P to the original vector V (see cited proposition for notation) instead of subtracting it, which wouldn't make the resulting vector orthogonal to the defined span $[\sin(x), \sin(3x)]$ (trust).
So to make $g(x)$ orthogonal to the span, the sign has to be cancelled out, this can be done by 'reversing' the constants c_1 and c_3 from equation(10)(11), note the sin() isn't part of the constants.

$$\frac{4}{\pi} + \frac{4}{3\pi} \to (-\frac{4}{\pi}) - \frac{4}{3\pi} \tag{13}$$

this should cancel out the '+' sign in $g(x)$ and make it orthogonal, which can be shown by evaluating the inner product $\langle g(x), \sin(x)\rangle$.

$$\langle g(x), \sin(x)\rangle = \int_0^\pi (1 + c_1\sin(x) + c_3\sin(3x)) \cdot \sin(x)\, dx$$

Looking inside the integral, it can be simplifed to:

$$(1 + c_1\sin(x) + c_3\sin(3x)) \cdot \sin(x) \to \sin(x) + c_1\sin^2(x) + c_3\sin(3x)\sin(x)$$

and from earlier proofs, we ascertain $\sin(x)\sin(3x) = 0$.

$$\int_0^\pi \sin(x) + c_1\sin^2(x)\, dx$$

to furter simplify

$$\int_0^\pi \sin(x)\,dx + c_1 \int_0^\pi \sin^2(x)\,dx$$

and from equation(9) $\int_0^\pi \sin^2(x)\,dx$ can be written as $\frac{\pi}{2}$.

$$\int_0^\pi \sin(x)\,dx + c_1 \frac{\pi}{2}$$

this can go even further beyond! looking back at the **Constant function** section, where for $k = 1$ then $\int_0^\pi \sin(x)\,dx = 2$

$$2 + c_1 \frac{\pi}{2}$$

then inserting c_1 from equation(13)

$$2 - \frac{4}{\pi} \cdot \frac{\pi}{2} = 2 - \frac{4\pi}{2\pi} = 0$$

proving that $g(x)$ is orthogonal to $\sin(x)$ and $\sin(3x)$, when making c_1 and c_3 negative.
To furher prove it's orthogonal for $\sin(2x)$ and $\sin(4x)$ to satisfy $k = 4$, note that $g(x)$ is a vector sum consisting of odd sine waves, and in the **Showing Orthogonality** section it was proved that odd sine waves are orthogonal to even sine waves, hence $g(x)$ must be orthogonal to all function in (2) for $k = 4$

References

[1] Andrew Swan. *Numerisk Lineær Algebra*. Institut for Matematik, Aarhus Universitet, 2025.